Experiment 4: the moment of inertia of an iOLab, from torque and from energy

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Uploaded to DTU Learn by Anastasios Tatarakis for the group.
Dummy hand in for practice. All data are simulated.

Purpose

Engineers who test scale model windmills fix an iOLab to the wings, so the iOLab adds its own moment of inertia to the rotating wings. To correct their measurements they need the moment of inertia of the iOLab about its y axis. We determine it in two independent ways, from the rotational form of Newton's second law and from energy conservation, by rolling the iOLab with a tape roll on each end down a ramp while its gyroscope measures the angular velocity.

1 Setup and measurements

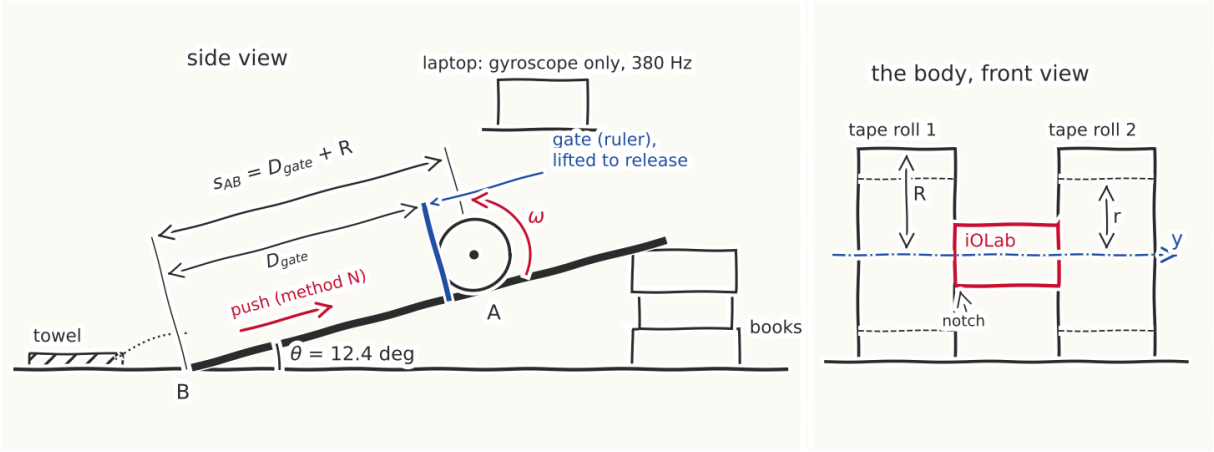


Figure 1. Hand drawn sketch of the setup. Left: the ramp seen from the side, angle drawn larger than real. Right: the body seen from the front; the tape rolls sit on the ends of the iOLab up to the notches, so the body turns about the y axis of the iOLab.

A tape roll was pressed onto each end of the iOLab up to the notches (Figure 1). The body rolls on the outer surface of the rolls and turns about the y axis of the iOLab; the gyroscope (only sensor, 380 Hz, range ± 34.9 rad/s) measured ω_y. A straight 0.90 m board rested with its lower end on the table and its upper end on books. ω and α count positive when the body rolls down. Table 1 lists the measured inputs.

Table 1. Measured inputs. R and k come from the calibration roll, $\sin\theta$ from the accelerometer (Section 3).

Quantity	Value	How	Quantity	Value	How
m_{iOLab}	(0.1962 ± 0.0001) kg	lab scale	R	(53.26 ± 0.06) mm	3 turns, 3 times
m_1, m_2	0.1842, 0.1878 ($\pm$0.0001) kg	lab scale	k	1.0113 ± 0.0009	3 turns, 3 times
M	(0.5682 ± 0.0001) kg	whole body	$\sin\theta$	0.2149 ± 0.0003	flips, 3 places
$r_1 = r_2$	(38.10 ± 0.25) mm	caliper	θ	$(12.41 \pm 0.02)^\circ$	
R_1, R_2	53.25, 53.36 ($\pm$0.16) mm	paper strip	D_{gate}	(0.4470 ± 0.0010) m	tape measure
$w \times t$	75.2×30.1 mm	ruler	I_{rolls}	$(7.99 \pm 0.03) \times 10^{-4}$ kg m^2	eq. (3)

2 Theory

Without slipping, $v = \omega R$ and $K = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}I_P\omega^2$, where I is the moment of inertia of the whole body about its axis and $I_P = I + MR^2$ the one about the contact line P (parallel axis theorem).

Torque (Method N). About P, only $Mg\sin\theta$ has a lever arm, R. Rolling resistance F_r opposes the motion: it acts with gravity when the body rolls up and against it when it rolls down, so $I_P\alpha_{\text{up}} = MgR\sin\theta + F_rR$ and $I_P\alpha_{\text{down}} = MgR\sin\theta - F_rR$. Adding and subtracting:

$$I_P = \frac{MgR\sin\theta}{\bar{\alpha}}, \qquad \bar{\alpha} = \tfrac{1}{2}(\alpha_{\text{up}} + \alpha_{\text{down}}), \qquad \varepsilon = \frac{F_r}{Mg\sin\theta} = \frac{\alpha_{\text{up}} - \alpha_{\text{down}}}{\alpha_{\text{up}} + \alpha_{\text{down}}}. \tag{1}$$

Energy (Method E). From rest the body rolls s_{AB} along the board and drops $\Delta h = s_{AB}\sin\theta$. Rolling resistance and air drag take the fractions ε and Δf of $Mg\Delta h$:

$$Mg\Delta h = \tfrac{1}{2}I_P\,\omega_B^2 + (\varepsilon + \Delta f)\,Mg\Delta h \quad\Longrightarrow\quad I_P = \frac{2Mg\,s_{AB}\sin\theta\,(1 - \varepsilon - \Delta f)}{\omega_B^2}. \tag{2}$$

The iOLab. Each tape roll is a thick hollow cylinder, and the iOLab is what is left:

$$I_j = \tfrac{1}{2}m_j\left(r_j^2 + R_j^2\right), \qquad I_{\text{iOLab}} = I_P - MR^2 - I_{\text{rolls}}, \quad I_{\text{rolls}} = I_1 + I_2. \tag{3}$$

Expectation. Seen along y the iOLab is a $w \times t$ box: $\frac{1}{12}m(w^2+t^2) = 1.07 \times 10^{-4}\,\mathrm{kg\,m^2}$ with its mass spread evenly, $\frac{1}{12}m(w+t)^2 = 1.81 \times 10^{-4}\,\mathrm{kg\,m^2}$ with all mass in its walls. The real value should lie between.

3 Method

Calibration roll. We rolled the body by hand along the table through exactly 3 turns, from a pen line on one roll touching a mark on the table to the same line touching it again, three times. The gyroscope angle $\Phi = \int \omega\,dt$ gave $k = \Phi/6\pi = 1.0113 \pm 0.0009$: the gyroscope reads 1.1 % too high, so we divided every gyroscope value by k. The distance between the marks gave $R = d/6\pi$, which agrees with R from the circumference, (53.31 ± 0.11) mm ($z = 0.4$).

Angle. The bare iOLab lay on its back on the board with y up the slope and then turned around, at three places; held upright with y up and then down, it gave a reference. $\sin\theta = (A_{y,\mathrm{up}} - A_{y,\mathrm{down}})/(A_{y\uparrow} - A_{y\downarrow})$ cancels the offset and the scale error of the axis. A ruler gave $(12.18 \pm 0.05)^\circ$ against the table top ($z = 4.1$ against the accelerometer). The same flip on the table showed a table tilt of $(0.226 \pm 0.004)^\circ$; with it the ruler gives $(12.41 \pm 0.05)^\circ$ ($z = 0.1$).

Method N, 7 runs: pushed up from 10 cm above the end, the body rolled 30 to 50 cm up and back and was caught. Straight lines fitted to $\omega(t)$ in two windows that skip the push, the turning point and the catch give α_{up} and α_{down} (Figure 2, left). **Method E**, 7 runs: the body rested against a ruler held across the board and started from rest when we lifted it. The gate face was D_{gate} from the end, so the contact line started $s_{AB} = D_{\mathrm{gate}} + R = (0.5003 \pm 0.0010)$ m from the end. At the end the body leaves the board and flies at constant ω, so $\omega(t)$ has a sharp corner; a fit of a straight rise followed by a flat part gives ω_B at the corner (Figure 2, right).

Analysis. The exported CSV files were analysed in Python. Uncertainties: $\sigma/\sqrt{N}$ for the mean of a still part, the standard error of fitted slopes, the standard deviation of the mean (SDOM) over runs, propagation with partial derivatives for independent inputs, and the standard score z. Air drag: $F_d = \frac{1}{2}\rho C_D A v^2$ with $C_D = 1.0 \pm 0.5$.

4 Results

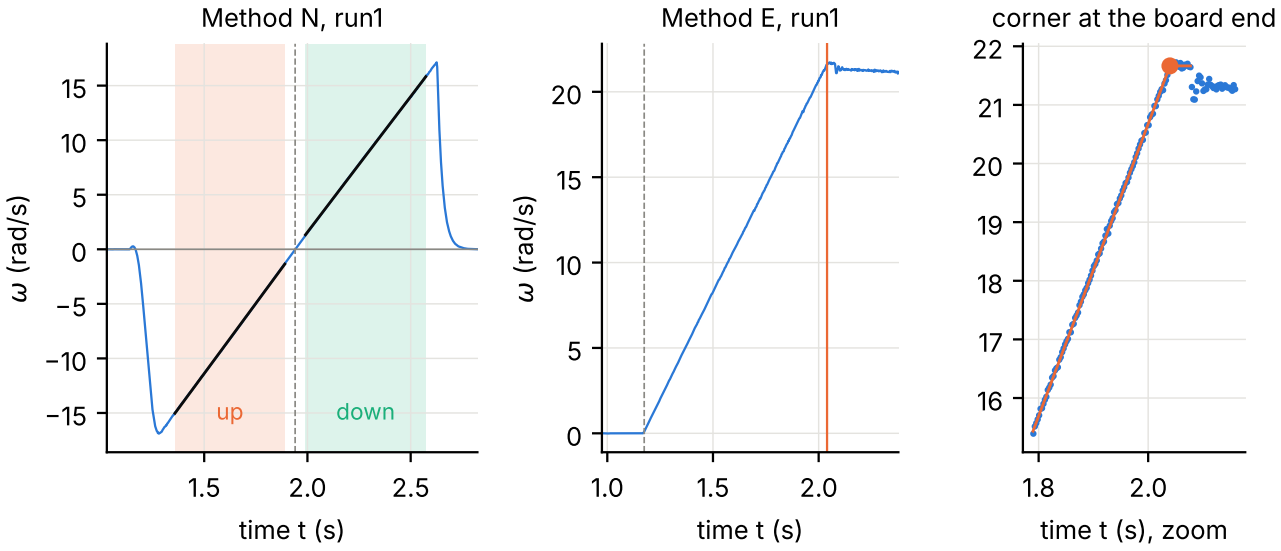


Figure 2. Left: Method N, run 1. After the push, ω is a straight line through zero; the slopes fitted in the up window and the down window are α_{up} and α_{down}. Middle and right: Method E, run 1, with a zoom on the corner where the body leaves the board, (t_B, ω_B), the short flight at constant ω and the landing.

Table 2. All runs, after dividing by k. The per run I_{iOLab} shows the scatter; the final values come from the means and the full uncertainty budget. Last row: mean ± SDOM.

	Method N				Method E		
Run	α_{up} (rad/s^2)	α_{down} (rad/s^2)	ε	I_{iOLab} (10^{-4} kg m^2)	Run	ω_B (rad/s)	I_{iOLab} (10^{-4} kg m^2)
1	25.410	24.583	0.0165	1.45	1	21.423	1.51
2	25.658	24.773	0.0176	1.23	2	21.445	1.46
3	25.477	24.645	0.0166	1.38	3	21.478	1.38
4	25.364	24.530	0.0167	1.50	4	21.455	1.44
5	25.571	24.722	0.0169	1.30	5	21.474	1.39
6	25.294	24.485	0.0163	1.56	6	21.447	1.45
7	25.566	24.706	0.0171	1.31	7	21.478	1.38
mean	25.48 ± 0.05	24.63 ± 0.04	0.0168 ± 0.0002		mean	21.457 ± 0.008	

Method N. $\bar{\alpha} = (25.06 \pm 0.04)\,\mathrm{rad/s^2}$ gives $I_P = (2.550 \pm 0.007) \times 10^{-3}\,\mathrm{kg\,m^2}$, $I = I_P - MR^2 = (9.38 \pm 0.06) \times 10^{-4}\,\mathrm{kg\,m^2}$ and $I_{\mathrm{iOLab}} = (1.39 \pm 0.10) \times 10^{-4}\,\mathrm{kg\,m^2}$. The friction fraction $\varepsilon = 0.0168 \pm 0.0002$ means a rolling

resistance of $F_r = (0.0202 \pm 0.0002)$ N.

Method E. $\omega_B = (21.457 \pm 0.008)$ rad/s, with ε from Method N and the drag correction $\Delta f = 0.0034 \pm 0.0017$, gives $I_P = (2.554 \pm 0.008) \times 10^{-3}$ kg m^2, $I = (9.42 \pm 0.10) \times 10^{-4}$ kg m^2 and $I_{\mathrm{iOLab}} = (1.43 \pm 0.12) \times 10^{-4}$ kg m^2. Checks: R times the gyroscope angle from release to the corner is (0.4986 ± 0.0004) m, against $s_{AB} = (0.5003 \pm 0.0010)$ m ($z = 1.4$), so the body did not slip. The slope of the release rise, (24.531 ± 0.003) rad/s^2, lies just below α_{down} ($z = 2.6$): the release runs are faster and lose a little more to air drag.

5 Discussion and conclusion

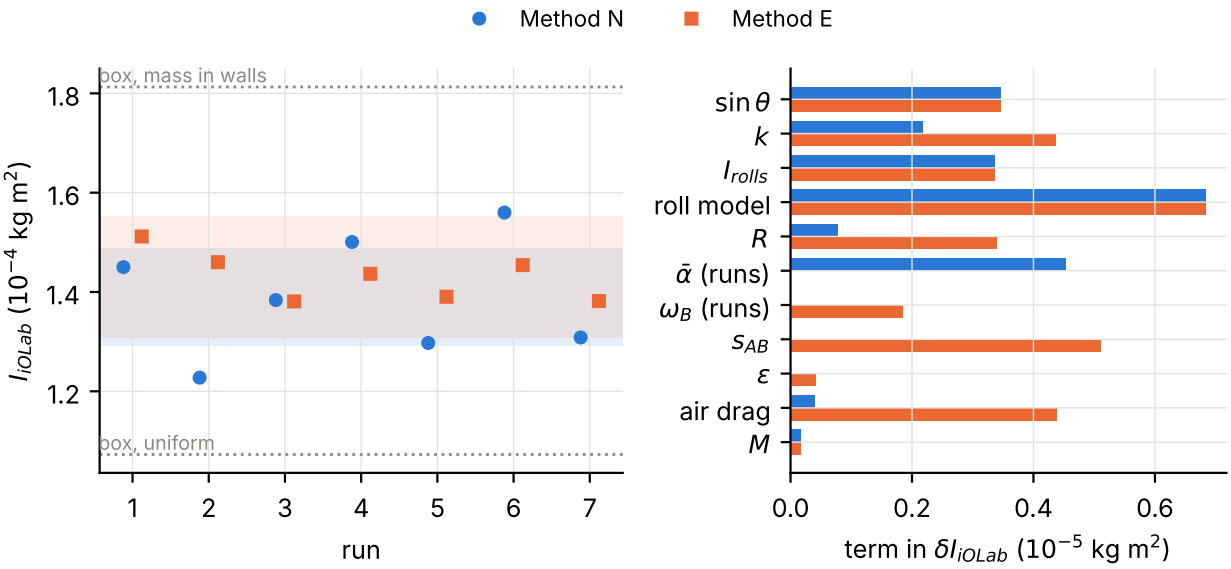


Figure 3. Left: I_{iOLab} from every run; the bands show each final value with its full uncertainty, the dotted lines the two box model limits. Right: the terms of the uncertainty budget, $|\partial I_{\mathrm{iOLab}}/\partial x|\,\delta x$, for both methods.

The two methods agree: $I_N - I_E = (-0.04 \pm 0.09) \times 10^{-4}$ kg m^2, $z = 0.5$. This uncertainty of the difference leaves out M, $\sin\theta$ and I_{rolls}, since an error in them moves both results by the same amount. Weighted by their independent parts, the combined result is

$$I_{\mathrm{iOLab}} = (1.40 \pm 0.10) \times 10^{-4}\,\mathrm{kg\,m^2},$$

inside the box model range 1.07×10^{-4} to 1.81×10^{-4} kg m^2 and closer to its lower end, as expected for an iOLab whose batteries and circuit board sit near the axis.

The iOLab is only 5.5 % of I_P; MR^2 makes up 63 % and the rolls 31 %. Every 1 % error in I_P is therefore an 18 % error in I_{iOLab}. Without the calibration roll, $k = 1.0113$ would have made Method N 20 % and Method E 40 % too small; without ε, Method E would be 31 % too large; the ruler angle without the table correction would have made both about a third too small. The budget (Figure 3, right) is led by the tape roll model: the cardboard core is lighter than tape, so (3) gives I_{rolls} about 0.9 % too small and I_{iOLab} up to 0.07×10^{-4} kg m^2 too large. Next come the run to run scatter of $\bar{\alpha}$, the angle, and in Method E the gate distance, the air drag estimate and k. Air drag makes Method E slightly too large; we corrected it and kept half of the correction as uncertainty.

For the windmill model, the iOLab adds $(1.40 \pm 0.10) \times 10^{-4}$ kg m^2 to the moment of inertia of the wings when its y axis lies on the axis of the wings; mounted a distance d from that axis, it adds $I_{\mathrm{iOLab}} + md^2$ (parallel axis theorem).

Reflections

1. How did you approach today's experiment: did you have expectations of the final result, or did you only get the result after you had the data?
Before the lab we estimated I_{iOLab} with a box model: between 1.1×10^{-4} and $1.8 \times 10^{-4}\,\text{kg}\,\text{m}^2$. We also worked out that the iOLab would be only about 5 % of what we actually measure, so we planned the calibration roll and the angle measurement before the first run. The value $1.40 \times 10^{-4}\,\text{kg}\,\text{m}^2$ came from the data, inside the range we expected. That the two methods would agree was not given in advance.

2. What did you learn from today's experiment?
A small difference of large numbers needs very precise inputs: 1 % in the angle is 18 % in the answer. A sensor can be checked against a motion we know exactly, three full turns, and ours read 1.1 % too high. Going up and coming down cancels friction, as in Experiment 3. Measuring the angle against gravity showed that our table was tilted by 0.2°, which a ruler alone would never have shown.

3. What do you think you were supposed to learn from today's experiment?
To apply the rotational form of Newton's second law and energy conservation to a rolling rigid body, with the parallel axis theorem, and to get the same quantity in two independent ways. Also to carry uncertainties through a calculation whose result is a small difference, and to find out which input limits the result.